

Stat 1080 Syllabus

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Office Hours By appointment. Schedule at https://calendly.com/vanderplas-stat/office-hours	Final Dec 16, 12:30-2:20

...the time may not be very remote when it will be understood that for complete initiation as an efficient citizen ... it is as necessary to be able to compute, to think in averages and maxima and minima, as it is now to be able to read and write.

– H.G. Wells, Mankind in the Making

Critical Information

Textbooks are free/open/online

- [Introduction to Data Science 1](#), by Rafael A. Irizarry (IDS 1)
- [Introduction to Data Science 2](#), by Rafael A. Irizarry (IDS 2)
- [Statistical Computing in R and Python](#) (SC).

Required Equipment & Software (Section [1.2](#)):

- A laptop that can run R (<https://r-project.org>, version $\geq 4.5.0$) and RStudio, OR a chrome-book that can access the web RStudio platform.
- Software packages up to date as of the beginning of the semester

Policy Summary (Section [3](#)):

- **Grade:** 20% Preparation & Participation, 20% Attendance, 40% Homework and Labs, 20% Project (Section [3.1](#)).
- **Attendance** matters. You get three excuse-free absences (Section [3.4.1](#)).
- **Prepare:** Reading (and reading quiz) must be done before class. (Section [3.5](#))
- Keep up with the work as assigned (Section [3.6](#)).
- **Communicate** early and often (Section [3.2](#)), especially if you're having trouble with something. Include the right details (Section [3.2.1](#)) when asking for help.
- Do not plan to travel during Finals Week – you must be present for project presentations (Section [3.8](#)).
- **AI use must be disclosed**, cited, with the full transcript and a diff between AI output and your submission. Be prepared to explain your thought process via oral exam (Section [3.10](#)).

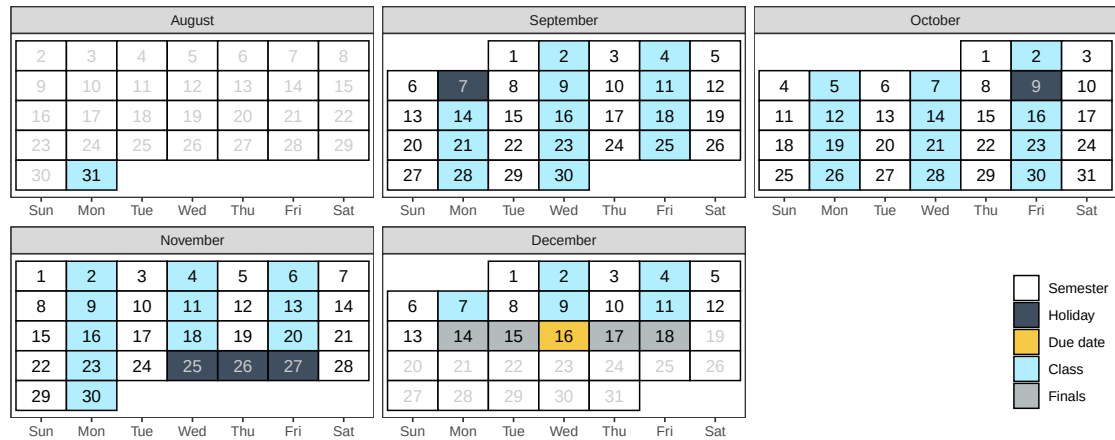


Figure 1: Course Calendar

1 Course Description

We live in a world with unprecedented access to information. Every day, our phones, computers, and “smart” devices collect mountains of data that are analyzed to inform and influence our choices, improve our health, and automate daily services. It is imperative that we understand how information is collected, visualized, and analyzed to shape our world. By learning about the life cycle of data, we can ensure that data is analyzed in productive and ethical ways.

This course provides your first introduction to data science. It assumes no prior experience. An introduction to data science requires learning key tools and principles from mathematics, statistics, and computer science. This course touches on all three.

1.1 Objectives

After taking this class, students should be able to:

1. Use computer programming to manipulate and visualize data.
2. Articulate insights from data in ways that answer existing questions and spark new ones.
3. Understand the concepts of randomness and statistical significance, and demonstrate both via computer simulation.
4. Use computer programming to develop data models that can predict quantities of interest.

Whether you see yourself as a future statistician, programmer, or simply want to make sense of the data-driven world around you, this course will give you tools to ask better questions, make informed decisions, and see the world in a new way.

1.2 Software

This course will involve programming in the R statistical programming language. R is a free and open-source language: a large community of people contribute to R without pay, and a

larger community of people use R without having to pay for it. The language is particularly good at data wrangling and data visualization. In addition, the community has developed a large number of extra packages to augment the language, making it easy to read in, visualize, and model new and different types of data.

Some of these packages also allow you to run other programming languages, such as Python, within R. This makes R a good, but not the only, platform to begin to learn programming essentials in a software actively used by professionals in the workforce.

You must have either a working laptop or a chromebook during each class period. **It will not be feasible to complete the programming assignments in this class on your phones.**

Laptop setup: You will need

- a [current version of R](#) (4.5.0 or later) installed
- [RStudio](#) (2026.1.0 or later) installed
- [Quarto](#) (1.10.0 or later) installed
- Updated versions of all software packages (as of the first week of class)

Chromebook setup: There is a browser based version of R and RStudio that you will be expected to access. Access is free for the first 25 compute hours. However, if your compute hours exceed the free allotment, you may need to upgrade to a student account for \$5/month. I recommend waiting to upgrade your account until you absolutely have to.

If your laptop breaks unexpectedly during the semester, contact the [IT service desk](#) for help. [Loaner laptops](#) may be available for short or long term use; you can then use the chromebook instructions to access RStudio in a browser until your laptop is fixed.

2 Tentative Schedule

This schedule is **tentative** and **subject to change** as the semester progresses. Students are expected to read the corresponding textbook chapter(s) (linked on the course website) **before coming to class**.

Week	Topic	Important Dates
1	Introduction	
2	Programming and RStudio	
3	Correlation is not Causation	
4	Programming in R	
5	The Tidyverse	
6	Dates, Times, and Distributions	
7	Data Visualization	
8	Statistics and Probability	
9	Data and Probability	
9	Random Variables	
10	Parameters and Estimates	
11	Statistical Inference	
12	Regression	
13	Regression	

Week	Topic	Important Dates
14	Project Meetings	
15	Finals Week	Project Presentations (12-16)

3 Course Policies

3.1 Assessment/Grading

Assignments	Weight
Preparation & Participation	20%
Attendance	20%
Personal Data Project	20%
Homework & Labs	40%

Lower bounds for grade cutoffs are shown in the following table. I will not “round up” grades at the end of the semester. I may, however, adjust this scale upwards, but this adjustment will be for the entire class.

Letter grade	X +	X	X -
A		93.0	90.0
B	87.0	83.0	80.0
C	77.0	73.0	70.0
D	67.0	63.0	60.0
F	<60.0		

Interpretation of this table: A grade of 92.4 will receive an A-. A grade of 86.3 will receive a B. A grade of 77.6 will receive a C+. A grade of 73.9 will receive a C. Anything below a 60 will receive an F.

3.2 Expectations

You can expect me to:

- reply to emails within 48 hours during the week (72 hours on weekends)
- be available in class to assist with assignments
- be available by appointment (office hours) for additional help or discussion

I expect you to:

- Be respectful and considerate of everyone in the class
- Engage in class discussions with civility and in good faith
- Complete the assigned reading, reading quiz, and any other activities before class
- Engage with the material and your classmates during class
- Seek help when you do not understand the material
- Communicate promptly if you anticipate that you will have trouble meeting deadlines or participating in a portion of the course.
- Do your own troubleshooting before contacting me for help (and mention things you’ve already tried when you do ask for help!)

Discussion and disagreement are important parts of the learning process, but it is important that mutual respect prevail. Individuals who detract from an atmosphere of civility and respect will be removed from the conversation and/or the classroom as applicable.

3.2.1 Technical Issues

If you have technical problems with R, quarto/rmarkdown, or RStudio send me an email with:

- any files necessary to replicate the issue
- a description of what you have tried, and
- a description of what you think should happen
- as much detail as possible:
 - Text copied (CTRL-C) from error messages
 - screenshots (not cell phone pics)

If you have problems with your computer, operating system, etc. that are not specific to this class, please consult IT/HelpDesk first.

3.3 Preparation & Participation

All students are expected to prepare for, attend and fully participate in class. Preparation & participation grades will be determined based on a combination of before-class reading checks (preparation) and in-class activities (participation).

Some in-class activities may be finished at home before they are submitted.

These assignments will typically be due at the end of the day and will close 24 hours after that. Assignments submitted during the 24-hour grace period will be graded.

3.4 Attendance

During each class, I will give you a numerical code which can be entered into a Canvas quiz to show that you were in class. The quiz will be open only during the class period, so you will need to have a device capable of interfacing with Canvas during class.

Sharing this code with anyone, or receiving a code for a class you were not present for, will be considered a violation of academic integrity.

Three absences (1 week of class) will be allowed without any reduction in your attendance score - no excuse is necessary! Please *do not email me* when you use your excuse-free absences - Canvas will automatically drop the quizzes associated with the first three missed classes.

3.4.1 Illness

If you are feeling ill, please **do not come to class**. Instead, review the material and work on the homework assignment, and then schedule an appointment with me to meet virtually. In the appointment reason field on [Calendly](#), indicate that this appointment is to substitute for your in-class participation on the date you missed.

If you need to miss more than 3 classes for illness, I will ask you to work with the Disability Resource Center in order to continue using this substitute for attendance.

3.4.2 Extenuating Circumstances

I am willing to offer some flexibility in extenuating circumstances if you let me know what is going on and how I can help. This flexibility takes effect *after* using your free absences, but

can also extend beyond attendance if appropriate.

In some situations, you may get accommodations only after working with the Disability Resource Center to document the problem.

Examples:

- Your dog had a seizure 20 minutes before an exam, and you were late because you had to get the dog to the vet.
 - Action: When getting the test, ask to talk for a minute. Explain the situation. Offer to provide vet records, if available.
 - Flexibility: Take a make-up exam, or start the exam immediately but get a few extra minutes to finish it.
- You have chronic migraines and have missed 8 classes this semester.
 - Action: Go to the DRC and get an accommodation for flexible attendance requirements.
 - Flexibility: Additional appointments to substitute for attendance beyond the 3 provided to all students.
- A train derailed, blocking your access to campus. You missed your scheduled presentation time.
 - Action: Email as soon as you know there is going to be a delay to explain the situation. If you are able, take pictures to document and attach to the email. (Note: only for situations, not illness or injury, please!) Attempt to find an alternate route to campus.
 - Flexibility: Present during a different time, or record a presentation.
 - Note: Flexibility here may be conditional on my ability to verify the situation.

3.5 Reading Checks

You will have weekly reading assignments which introduce the weekly focus area and provide different perspectives on the material. Some of these perspectives will be covered in class lectures and activities, but some will not be covered explicitly and are provided to expose you to different approaches and opinions.

Readings are linked on the course webpage for each week.

A reading check will be due on Canvas before class when there is reading assigned. This serves as a low-stakes, grade-based reminder to prepare for class. The lowest three reading checks will be dropped.

3.6 Homework

Several formative assignments (assignments meant to help you practice skills) will be given throughout the semester.

- Assignments must be submitted in the file format specified, and should run or compile as submitted. Code should be formatted using the style guidelines provided in class and demonstrated in videos posted to Canvas. See the homework [submission checklist](#) for more details.
- I will aim to grade assignments within a week of the specified due date.

3.7 Labs

Labs are class sessions that are dedicated to working interactively on an assignment. A lab assignment may need to be finished at home before it is submitted. Lab assignments should be checked using the [submission checklist](#) before they are submitted, as the same formatting standards will apply.

3.8 Project

You will complete a personal data collection project in this course. The project will be scaffolded - you will need to submit a proposal, as well as several intermediate assignments throughout the course of the semester, applying information we are covering in class to your own data. These intermediate submissions will contribute to your overall project grade.

3.8.1 Project Presentation Attendance

DO NOT make travel plans that assume you will not have to attend the final exam period. Personal data collection project speed presentations will take place during the final exam period.

3.9 Work Submission Policies

These policies apply to labs, homework assignments, and project components.

- Assignments submitted by email or through a means other than Canvas will receive a 0.
- Late assignments will be accepted only under extenuating circumstances, and only if you have contacted me **prior** to the assignment due date and received permission to hand the assignment in late.
DO NOT email me after the assignment close date or due date asking for permission to submit assignments late. Permission to turn an assignment in late must be obtained **before** the due date.
- **Logistics:** Assignments are due at 12:59 pm on Fridays. There is a grace period in Canvas between the due date and the assignment close date. After the close date, you will not be able to submit your assignment in Canvas.
- I reserve the right not to grade (or to assign a 0 to) any assignments received after the assignment due date. However, in practice, assignments received before the assignment close date will typically be graded.

3.10 Evaluation and Academic Integrity

Students are expected to adhere to guidelines concerning academic dishonesty outlined in [Article VI University Regulations Regarding Academic Integrity](#). In this course, academic dishonesty includes failure to disclose use of AI (depending on sources of aid beyond those permitted), as well as more traditional examples of academic dishonesty.

The policies detailed below stand in addition to university policies.

Summary: I may, at my discretion, give students an oral exam on their submitted assignment to replace the graded assignment itself. That is, if you submitted it, you should be able to defend, discuss, and justify your work.

3.10.1 Evaluation Criteria

In every assignment, discussion, and written component of this class, you are expected to **demonstrate that you are intellectually engaging with the material**. I will evaluate you based on this engagement, which means that technically correct but low effort answers which do not demonstrate engagement or understanding will receive no credit.

When you answer questions in this class, your goal is to show that you either understand the material or are actively engaging with it. If you did not achieve this goal, then your answer is incomplete, regardless of whether or not it is technically correct. This is not to encourage you to add unnecessary complexity to your answer - simple, elegant solutions are always preferable to unwieldy, complex solutions that accomplish the same task.

Grammar and spelling are important, as is your ability to communicate technical information clearly in a written format; both of these criteria will be used **in addition to assignment-specific rubrics** to evaluate your work.

3.10.2 Code

You must be able to explain how the logic works for any code you turn in. This means that code you obtained from e.g. StackOverflow is fine to use if you can explain it and modify it for the purposes of this class, but if you cannot explain your code you will not get credit for the assignment. This is in line with what is generally considered acceptable behavior in programming - reuse is fine (subject to the code's license) but you must be able to fully explain and modify any code you did not write yourself. You should also provide a link to the source of your code (or even the logic behind your code) - this is useful both for my information and for future you, in addition to being a good way to give attribution to the original author.

3.10.3 Oral Exam Policy

With the proliferation of AI large language models, I reserve the right to replace homework grades with grades based on an oral discussion of your submissions. If you cannot explain the logic behind your submission, how you came to the approach you used, and identify the tradeoffs behind decisions you made, then you will not receive credit. LLMs can be a useful tool, but this course's objectives are to develop your understanding of data science, not your ability to use LLMs. Fundamentally, you need to understand concepts and apply methodology in order to use data effectively; generative AI solutions may hamper both of these goals if you rely on them too much.

3.10.4 Generative AI

It may be useful to leverage AI tools to ensure that your work conforms to grammar and style guidelines, but I very highly discourage the use of generative AI to actually write content or to execute a data analysis. I highly prefer *your* writing and thought process, even with some mistakes, to the bland content generated by most LLMs.

Any use of generative AI must be disclosed in an appendix to your submission - this includes brainstorming, editing, using AI as spell-check/grammar-check, and so on. You must document the following:

- the version of the generative AI used (e.g. Claude Sonnet 4.5, GPT 5.2, Copilot GPT-5 chat)
- the full sequence of prompts and responses
- any additional inputs you provided to the AI system (documents, images, etc.)
- a “diff” between the AI responses and your submission, showing exactly what was generated by the AI system and what you changed. If you are using quarto, you can use the `diff` command line tool to get this (you may have to fix headers) after you’ve pasted the AI text into its own document.

4 University Syllabus Policies

See <https://www.usu.edu/teach/help-topics/teaching-tips/syllabus-resources> for campus policies including the following:

- Academic Support
- Classroom Behavior
- Academic Integrity and Dishonesty
- Discrimination and Misconduct
- Withdrawal and “I” grade policy
- Disability services and pregnancy accommodations
- Grievance Processes
- Emergency Procedures
- Mental Health and Food Security resources

4.1 Disability Resource Center (DRC)

USU welcomes students with disabilities. If you have, or suspect you may have, a physical, mental health, or learning disability that may require accommodations in this course, please contact the Disability Resource Center (DRC) as early in the semester as possible (University Inn #101, 435-797-2444, drc@usu.edu). All disability related accommodations must be approved by the DRC. Once approved, the DRC will coordinate with faculty to provide accommodations.